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# N + - Network Fundamentals & Building Networks.

Firewall Basics.

## TASK 1.

- ❖ **Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.**

### ➤ **Blocking Spotify Using Windows Defender Firewall:**

#### **Steps:**

- Open **Windows Defender Firewall with Advanced Security**.
- Select **Outbound Rules** from the left panel.
- Click **New Rule** and choose **Program**.
- Select the **Spotify desktop application (.exe)** and click **Next**.
- Choose **Block the connection** and continue.
- Apply the rule to **Domain, Private, and Public** profiles.
- Name the rule **Block Spotify** and click **Finish**.

**Result:** The outbound firewall rule blocks all network traffic from the Spotify desktop application.

## TASK 2.

- ❖ **Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).**

### ➤ **Inbound Firewall Rule for Google Chrome:**

#### **Steps:**

- Open **Windows Defender Firewall with Advanced Security**.
- Select **Inbound Rules** and click **New Rule**.
- Choose **Port** and select **TCP**.

- Enter **443** as the specific local port.
- Select **Allow the connection**.
- Apply the rule to the required network profiles.
- Name the rule **Allow Chrome HTTPS (443)** and click **Finish**.

**Result:** The inbound rule allows **HTTPS traffic on TCP port 443** for Google Chrome.

## TASK 3.

❖ **Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall, Use the 'Scope' tab when creating a new rule to specify the IP address.**

### ➤ **Blocking Traffic from a Specific IP Address:**

#### **Steps:**

- Open **Windows Defender Firewall with Advanced Security**.
- Select **Inbound Rules** and click **New Rule**.
- Choose **Custom** and click **Next**.
- Keep the default settings for **Program** and **Protocol and Ports**, then click **Next**.
- In the **Scope** tab, under **Remote IP address**, select **These IP addresses**.
- Click **Add** and enter **123.45.67.89**, then click **Next**.
- Select **Block the connection** and click **Next**.
- Apply the rule to the required network profiles.
- Name the rule **Block 123.45.67.89** and click **Finish**.

**Result:** All incoming traffic from **123.45.67.89** is blocked by Windows Defender Firewall.

## TASK 4.

- ❖ Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

➤ **Hardware Firewall vs Software Firewall:**

| Point of Difference | Hardware Firewall                                                                        | Software Firewall                                                                          |
|---------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Location            | A separate physical device placed between the network and the Internet.                  | A program installed and running on an individual computer.                                 |
| Protection          | Protects multiple devices on the entire network.                                         | Mainly protects the device on which it is installed.                                       |
| Performance         | Handles network traffic efficiently without using the resources of individual computers. | Uses some CPU and memory of the computer.                                                  |
| Use Case            | Large office: Protects the company network from unauthorized Internet traffic.           | Windows PC: Windows Defender Firewall blocks unwanted connections to protect the computer. |